Chris Zhengyu Liu

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EDUCATION

Georgia Institute of Technology — Expected Spring 2028
B.S. Industrial & Systems Engineering (GPA: 4.0/4.0) — *Atlanta, GA*

- Concentration: Advanced Studies for Operations Research & Statistics | Minor: Computing & Artificial Intelligence
- Coursework: Data Structures & Algorithms, Database Systems, Probability, Stochastic Processes, Linear Algebra

EXPERIENCE

School of Civil & Environmental Engineering – Georgia Tech — Jan 2026–Present
Programming Research Assistant – Research Software & Machine Vision — *Atlanta, GA*

- Own development of a Flask/REST and JavaScript dashboard for a sidewalk simulation platform modeling 18 mobility profiles; replaced notebook-based setup with configurable networks, accessibility cost add-ons, and batch execution.
- Designed multithreaded job-queue and scenario-staging workflows to run, monitor, and aggregate tens of thousands of simulations with progress tracking and geospatial shapefile exports.
- Designed database schema, add-on architecture, and playback workflows for pedestrian-asset management tools while collaborating across a 20-member interdisciplinary research group.

Scheller College of Business – Georgia Tech — Jun 2024–May 2025
Research Assistant – Data Engineering & Automation (Dr. Tracey Swartz) — *Atlanta, GA*

- Built a Python/Linux ETL pipeline for a 17 GB corpus spanning 10,000+ filings, shareholder letters, and executive web profiles using Selenium, PyMuPDF, OCR, XML, Regex, Pandas, and multiprocessing.
- Replaced a months-long manual workflow with parallelized collection and extraction, exporting research-ready Excel datasets and routing ambiguous records for human review.
- Engineered TF-IDF features from shareholder communications and authored protocols used to train and coordinate two interns on web scraping and data processing.

PROJECTS

desmos.py / desmosdotpy – GitHub | PyPI — May 2026–Present

- Published an open-source Python package that compiles a subset of Python AST into Desmos LaTeX/state JSON and emits self-contained interactive HTML graphs.
- Implemented vectorized math translation, geometry, controls, and line-specific errors; validated the compiler with 37 golden-file tests and built a roughly 40-line Mandelbrot explorer.

GT Bus Arrival Prediction Dashboard – Live Demo — Apr–May 2026

- Built an end-to-end transit data system that polls Georgia Tech's public API every 15 seconds, stores 1M+ GPS observations, derives observed next-stop arrivals, and trains an XGBoost ETA model.
- Deployed a Vultr-hosted web dashboard for live predictions and historical route analysis; independently owned collection, ETL, feature engineering, modeling, interface development, and deployment.

Classify – AI Classroom Platform — Jan 2026–Present

- Designed the Flask backend and Gemini integration for a time-synchronized infinite canvas that analyzes each step of student work and returns targeted feedback.
- Built hierarchical aggregation from step feedback to assignment and class summaries, persisted results in PostgreSQL, and deployed the prototype with AWS EC2 and Vercel.

Evolutionary Symbolic Regression for Crypto – GitHub — Aug 2025–May 2026

- Built a DEAP/CuPy genetic-programming system that evolves tree-based strategies across market-regime specialists; automated Binance ingestion, Parquet conversion, multi-timeframe indicators, and out-of-sample backtests.

TECHNICAL SKILLS

Languages: Python, Java, SQL
ML & Data: XGBoost, Scikit-learn, DEAP, CuPy, Pandas, NumPy, Geospatial Processing
Backend & Tools: Flask, REST APIs, MySQL, PostgreSQL, Selenium, Git, Linux, AWS EC2, GCP Compute Engine, PyPI

HONORS & ACTIVITIES

- Faculty Honors, Georgia Tech (Fall 2025, Spring 2026) | 3rd Place, HackGwinnett 2024
- Published Poet: Live Poets Society of NJ (Winter 2025, Topical Winner), Poetry Lab Shanghai (Spring 2025)